competishun — Power of real gurus **MATHEMATICS** **TARGET : JEE (Main) 2024**	**CHAMP-2024** **REVISION DPP** **NO. 19**

Syllabus :- VECTOR

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **22 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b}$ -is a vector such that $\vec{a} \cdot \vec{b} = 0$ and $\vec{a} \times \vec{b} = \vec{0}$. Then which of following is correct?

(1) $\vec{b} = 0$ (2) $\vec{b} \perp \vec{a}$

(3) $\vec{b}$ is non-zero vector (4) $\vec{b}$ is parallel to $\vec{a}$

2. $\vec{a}, \vec{b}, \vec{c}$ are three non-collinear vectors such that $\vec{a} + \vec{b}$ is parallel to $\vec{c}$ and $\vec{a} + \vec{c}$ is parallel to $\vec{b}$. Then

(1) $\vec{a} + \vec{b} = \vec{c}$ (2) $\vec{b} + \vec{c} = \vec{a}$ (3) $\vec{a} + \vec{c} - \vec{b} = \vec{0}$ (4) $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

3. $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors satisfying relation $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b}}{2}$ where $\vec{b}$ and $\vec{c}$ are non-parallel. The angle between vectors $\vec{a}$ and $\vec{b}$ can be

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{2}$

4. Let $\hat{a}, \hat{b}$ be unit vectors. If $\vec{c}$ be a vector such that the angle between $\hat{a}$ and $\hat{c}$ is $\frac{\pi}{12}$, and $\hat{b} = \vec{c} + 2(\vec{c} \times \hat{a})$, then $|6\vec{c}|^2$ is equal to

(1) $6(3 - \sqrt{3})$ (2) $3 + \sqrt{3}$ (3) $6(3 + \sqrt{3})$ (4) $6(\sqrt{3} + 1)$

5. The coordinates of foot of perpendicular drawn from point A(–2, 3, 0) on line PQ making equal angles with coordinate axes where point P is (–3, 5, 2), is

(1) (–4, 4, 1) (2) (4, 4, 1) (3) (–2, 2, 1) (4) (2, 2, 1)

6. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ & $\vec{a} \cdot \vec{b} = 1$ & $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$ then $\vec{b}$ is equal to -

(1) $2\hat{i}$ (2) $\hat{i} - \hat{j} + \hat{k}$ (3) $\hat{i}$ (4) $2\hat{j} - \hat{k}$

7. Let a, b, c be distinct non-negative numbers and the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$, $c\hat{i} + c\hat{j} + b\hat{k}$ lie in a plane, then the quadratic equation $ax^2 + 2cx + b = 0$ has -

(1) real and equal roots (2) real unequal roots

(3) unreal roots (4) both roots real and positive

8. If A(6,3,2), B(5,1,4) , C(3,–4,7), D(0,2,5) are four points then length of projection of CD on AB is

(1) $\frac{13}{3}$ (2) $\frac{13}{7}$ (3) $\frac{3}{13}$ (4) $\frac{7}{13}$

9. For any four points P, Q, R, S, $\left|\vec{PQ}\times\vec{RS} - \vec{QR}\times\vec{PS} + \vec{RP}\times\vec{QS}\right|$ is equal to 4 times the area of the triangle–

(1) PQR (2) QRS (3) PRS (4) PQS

10. A vector of magnitude 3, bisecting the angle between the vectors $\bar{a} = 2i + j - k$ and $\bar{b} = i - 2j + k$ and making an obtuse angle with $\bar{b}$ is –

(1) $\frac{3i-j}{\sqrt{6}}$ (2) $\frac{i+3j-2k}{\sqrt{14}}$ (3) $\frac{3(i+3j-2k)}{\sqrt{14}}$ (4) $\frac{3i-j}{\sqrt{10}}$

11. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that $\left|\left(\hat{a}+\hat{b}\right)+2(\hat{a}\times\hat{b})\right| = 2$. If $\theta \in (0, \pi)$ is the angle between $\hat{a}$ and $\hat{b}$, then among the statements :

(S1) : $2\left|\hat{a}\times\hat{b}\right| = \left|\hat{a}-\hat{b}\right|$

(S2) : The projection of $\hat{a}$ on $\left(\hat{a}+\hat{b}\right)$ is $\frac{1}{2}$

(1) Only (S1) is true (2) Only (S2) is true

(3) Both (S1) and (S2) are true (4) Both (S1) and (S2) are false

12. A unit vector is orthogonal to $5\hat{i} + 2\hat{j} + 6\hat{k}$ and is coplanar to $2\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ then the vector is -

(1) $\frac{3\hat{j}-\hat{k}}{\sqrt{10}}$ (2) $\frac{2\hat{i}+5\hat{j}}{\sqrt{29}}$ (3) $\frac{6\hat{i}-5\hat{k}}{\sqrt{61}}$ (4) $\frac{2\hat{i}+2\hat{j}-\hat{k}}{3}$

13. Let $\vec{a}, \vec{b}$ are the position vectors of A, B respectively and C is a point on AB produced such that AC = 3 AB. If D is harmonic conjugate of C with respect to A and B, then the position vector of D is

(1) $3\vec{a} - 2\vec{b}$ (2) $3\vec{b} - 2\vec{a}$ (3) $\frac{2\vec{b}+3\vec{a}}{5}$ (4) None of these

14. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ $a_i > 0$, i = 1, 2, 3 be a vector which makes equal angles with the coordinates axes OX, OY and OZ. Also, let the projection of $\vec{a}$ on the vector $3\hat{i}+4\hat{j}$ be 7. Let $\vec{b}$ be a vector obtained by rotating $\vec{a}$ with 90°. If $\vec{a},\vec{b}$ and x-axis are coplanar, then projection of a vector $\vec{b}$ on $3\hat{i}+4\hat{j}$ is equal to

(1) $\sqrt{7}$ (2) $\sqrt{2}$ (3) 2 (4) 7

15. If $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ & $\vec{b}=\hat{i}-2\hat{j}+\hat{k}$, then the vector $\vec{c}$ such that $\vec{a}\cdot\vec{c}=2$ & $\vec{a}\times\vec{c}=\vec{b}$ -is

(1) $\frac{1}{3}\left(\hat{i}-2\hat{j}+\hat{k}\right)$ (2) $\frac{1}{3}\left(-\hat{i}+2\hat{j}+5\hat{k}\right)$ (3) $\frac{1}{3}\left(\hat{i}+2\hat{j}-5\hat{k}\right)$ (4) $\frac{1}{3}\left(-\hat{i}+2\hat{j}-5\hat{k}\right)$

16. If $\vec{a}=\vec{i}+\vec{j}+\vec{k}$, $\vec{b}=4\vec{i}+3\vec{j}+4\vec{k}$ and $\vec{c}=\vec{i}+\alpha\vec{j}+\beta\vec{k}$ are linearly dependent vectors and $|\vec{c}|=\sqrt{3}$, then

(1) $\alpha=1, \beta=-1$ (2) $\alpha=1, \beta=\pm1$ (3) $\alpha=-1, \beta=\pm1$ (4) $\alpha=1$ $\beta=,1$

17. Let $\vec{a}$ and $\vec{b}$ be two non-collinear unit vectors. If $\vec{\mu}=\vec{a}-\left(\vec{a}.\vec{b}\right)\vec{b}$ and $\left(\vec{v}\right)=\vec{a}\times\vec{b}$, then $|\vec{v}|$ is equal to

(1) $|\vec{\mu}|-|\vec{\mu}\cdot\vec{a}|$ (2) $|\vec{\mu}|+|\vec{\mu}\cdot\vec{a}|$ (3) $|\vec{\mu}|+|\vec{\mu}\cdot\vec{b}|$ (4) $|\vec{\mu}|+\vec{\mu}\cdot\left(\vec{a}+\vec{b}\right)$

18. Let $\vec{\lambda}=\vec{a}\times(\vec{b}+\vec{c})$, $\vec{\mu}=\vec{b}\times(\vec{c}+\vec{a})$, and $\vec{v}=\vec{c}\times(\vec{a}+\vec{b})$, . Then

(1) $\vec{\lambda}+\vec{\mu}=\vec{v}$ (2) $\vec{\lambda}$, $\vec{\mu}$, $\vec{v}$ are coplanar

(3) $\vec{\lambda}+\vec{v}=2\vec{\mu}$ (4) None of these

19. Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$ where $|\vec{a}|$ = 4, $|\vec{b}|$ = 3 $\theta\in\left(\frac{\pi}{4},\frac{\pi}{3}\right)$. Then $\left|\vec{a}-\vec{b}\times\left(\vec{a}+\vec{b}\right)^2\right|+\times\left(\vec{a}.\vec{b}\right)^2$ is equal to ______.

(1) 575 (2) 574 (3) 576 (4) 576

20. Let α, β, γ be distinct real numbers . The points with position vectors $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$, $\beta\hat{i}+\gamma\hat{j}+\alpha\hat{k}$, $\gamma\hat{i}+\alpha\hat{j}+\beta\hat{k}$

(1) are collinear (2) form an equilateral triangle

(3) form a scalene triangle (4) form a right angled triangle

SECTION-II
(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. Given that $\vec{e}_1 = \vec{i} + \vec{j}$, $\vec{e}_2 = \vec{i} + \vec{k}$, $\vec{e}_3 = \vec{j} + \vec{k}$ and if $5\vec{i} + 2\vec{j} + \vec{k} = \ell_1\vec{e}_1 + \ell_2\vec{e}_2 + \ell_3\vec{e}_3$, then $(\ell_1+\ell_2+\ell_3)$ is equal to

22. Given that $\vec{a} = \alpha^2\vec{i} + \alpha\hat{j} + \hat{k}$ and $\vec{b} = \alpha\vec{i} - \alpha^2\hat{j} + \hat{k}$ are equally inclined to $\vec{c} = \hat{i} - 3\hat{j} + 4\vec{k}$, the angle of inclination is $\sin^{-1}\frac{x}{\sqrt{39}}$ find x

23. Given the statements :

S_1 : The sum of two unit vectors can be a unit vector

S_2 : The magnitude of difference between two unit vectors can be greater than the magnitude of a unit vector . How many of these statements are true :

24. Let $\vec{x} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{y} = \hat{i} + \hat{j}$. If $\vec{z}$ is a vector such that $\vec{x} \cdot \vec{z} = |\vec{z}|$, $|\vec{z} - \vec{x}| = 2\sqrt{2}$ and the angle betweeen $\vec{x} \times \vec{y}$ and $\vec{z}$ is 30°, then the magnitude of the vector $(\vec{x} \times \vec{y}) \times \vec{z}$ is:

25. Let $\vec{b} = \hat{i} + \hat{j} + \lambda\hat{k}$, $\lambda \in R$. If $\vec{a}$ is a vector such that $\vec{a} \times \vec{b} = 13\hat{i} - \hat{j} - 4\hat{k}$ and $\vec{a} \cdot \vec{b} + 21 = 0$, then $(\vec{b} - \vec{a}) \cdot (\hat{k} - \hat{j}) + (\vec{b} + \vec{a}) \cdot (\hat{i} - \hat{k})$ is equal to

Anwer Key_DPP-19

1.	**(1)**	**2.**	**(4)**	**3.**	**(4)**	**4.**	**(3)**	**5.**	**(1)**	**6.**	**(3)**	**7.**	**(1)**
8.	**(1)**	**9.**	**(2)**	**10.**	**(3)**	**11.**	**(3)**	**12.**	**(1)**	**13.**	**(4)**	**14.**	**(2)**
15.	**(2)**	**16.**	**(4)**	**17.**	**(3)**	**18.**	**(2)**	**19.**	**(4)**	**20.**	**(2)**	**21.**	**(4)**
22.	**(5)**	**23.**	**(2)**	**24.**	**(1.5)**	**25.**	**(14)**						